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Report
Milestone - 2

Dataset Generation & Machine Learning Model Development

TEAM 3
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1. Overview

The second milestone focuses on generating synthetic environmental data and developing a machine learning model for predicting health risk levels. This stage extends the system design from the previous milestone by incorporating data-driven decision-making and predictive analytics.

The primary objective is to simulate real-world environmental conditions, preprocess the dataset, classify risk levels, and train a model capable of accurately identifying unsafe conditions in enclosed environments.

2. Dataset Generation

Parameters Selected

The dataset was designed to represent key environmental factors affecting air quality and human health. The following parameters were considered:

- Temperature (°C)
- Humidity (%)
- Air Quality (PPM)
- Pressure (hPa)
- Timestamp

Justification of Parameters

- **Temperature** affects comfort and can influence air circulation.
- **Humidity** impacts respiratory conditions and air quality perception.
- **Air Quality (PPM)** is the primary indicator of pollution levels.
- **Pressure** provides additional environmental context and stability.

These parameters were selected based on their relevance in indoor air quality monitoring systems.

Data Range Selection

The values were generated within realistic ranges to simulate actual sensor behavior:

Parameter	Range	Reason
Temperature	18 – 40 °C	Indoor variation
Humidity	20 – 90 %	Dry to high humidity
Air Quality	100 – 600 PPM	Clean to polluted air
Pressure	980 – 1030 hPa	Normal atmospheric range

Dataset Creation

A total of 400 records were generated using Python, with each entry representing environmental conditions at regular time intervals. The dataset was stored in CSV format.

Additionally, a **l**arge-scale dataset of 200,000 records was generated to simulate real-world IoT data volume and demonstrate system scalability.

```csv
Temperature,Humidity,AirQuality,Pressure,Risk
27.49080237694725,46.95527802046611,450.0288777502409,980.1023718270944,High
39.01428612819832,57.520595430622095,192.07104112672047,990.1925122831316,High
34.6398788362281,35.95292987517835,73.35936401394224,1026.9572819937537,Moderate
31.973169683940732,56.81022152059061 6,463.8848148397874,1002.1863020482529,High
23.12037280884873,42.184880978698814,246.31735224691798,986.1163179287377,Moderate
23.119890406724053,46.65571824576245,102.57043884444171,1004.3896846061868,Safe
21.161672243363988,58.68675389668032,211.6467107852217,990.7645214274507,Moderate
37.3235229154987,83.17804068519783,401.976047173145,997.5969766163367,High
32.022300234864176,64.70443810314124,310.57797750057256,1026.6894208968445,High
34.161451555920905,31.060087542382885,473.35691735049096,1015.9730050275546,High
20.411689885916047,38.610697569002 6,494.57845385312066,1013.1183181637967,High
39.39819704323989,33.80053614787348,214.6637787582082 8,1018.4844930013709,High
36.648852816008436,41.175911100312874,446.9234737276733,1008.0639980082252,High
24.246782213565524,50.619841016790346,480.98827362964033,990.3518866137949,High
23.636499344142013,40.50369196010409,140.97192799166197,991.002262736499,Safe
23.668090197068675,57.700691625854994,460.2606649123879,1022.3775837389738,High
26.084844859190753,79.45102066485609,117.57865521146101,992.1587640649497,Moderate
30.495128632644757,79.53255348545562,284.836187017126 8,996.2730574390491,Moderate
28.638900372842315,50.377947084539095,413.8786866792198,980.0672776775775,High
25.824582803960837,49.600201560909 1,210.71781490821465,990.7979956667167,Moderate
32.23705789444759,74.745216858767 9,120.563994947584,1005.0067820285195,Moderate
22.789877213040835,54.90436878059161,194.47158579246312,990.3514694628677,Moderate
25.842892970704362,88.20696146966415,310.83393073601343,988.864763784716,High
27.327236865873836,56.697233694289785,217.35267106352504,1010.2818793784916,Moderate
29.12139968434072,44.59084966670035,397.0661267520981,1004.8157080428361,High
35.70351922786027,30.58921550003449,363.95281503113387,1023.4833452034769,High
23.993475643167194,87.24677233587545,380.0612397103573,1027.273865121069,High
30.284688768272233,37.12463038573861,324.00153283656374,987.7880290983911,High
31.848291377240848,40.51060087535291,94.80291697406861,1028.0057122865087,Moderate
20.929008254399953,32.30021867753507,481.37507065778766,1012.9784832456248,High
32.150897038028766,57.89624767640784,141.9574085390929,991.3031834017138,Moderate
23.41048247374583,54.50316106879052,51.235278632995524,1003.1937034819944,Safe
21.30103185970559,78.94598363203866,452.88032913121026,1004.646058634161,High
38.977710745066666,63.32150055725894,442.32032706475366,995.8908178174627,High
39.31264066149119,64.93761000575697,488.56132179642606,1014.6301528580857,High
36.167946962329225,77.41464785941584,224.14639845569633,991.4663077577165,High
26.092275383467413,64.97593813127806,481.47902218002514,1019.5483289393915,High
21.95344228012768,63.92274752923127,419.2841021938779,1012.2199040875491,High
33.684660530243136,85.71153837640836,361.8825713185548,1013.7021354203331,High
```

3. Data Preprocessing

The dataset was prepared for machine learning through the following steps:

- Structuring data into tabular format
- Ensuring consistent timestamp format
- Selecting relevant features
- Verifying data consistency

The preprocessing ensured that the dataset was clean, structured, and suitable for training machine learning models.

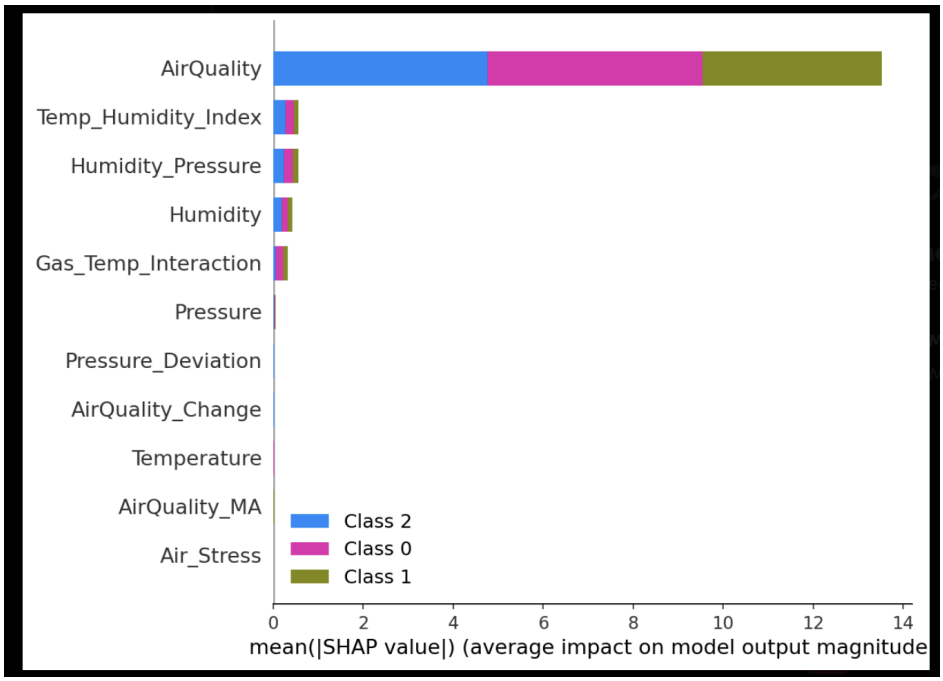
4. Risk Classification

A rule-based approach was used to label the dataset into three categories:

- Safe
- Moderate Risk
- High Risk

Classification Conditions

- High Risk
  - Air Quality > 400
  - Temperature < 20 or > 35
  - Humidity > 80
- Moderate Risk
  - Air Quality between 250 – 400
  - Moderate humidity variation
- Safe
  - Air Quality < 250
  - Normal temperature and humidity



5. Machine Learning Model Development

Model Selection

Initially, multiple machine learning approaches were considered, including:

- Random Forest
- Neural Networks
- XGBoost

The Random Forest Classifier was selected for implementation due to its balance between accuracy, simplicity, and computational efficiency.

Model Training Process

- Dataset split into **80% training and 20% testing**
- Features used:
  - Temperature
  - Humidity
  - Air Quality
  - Pressure
- Target variable:
  - Risk Level

The model was trained using the training dataset and evaluated using the testing dataset.

6. Model Comparison and Justification

Model	Advantages	Limitations	Suitability
Random Forest	High accuracy, simple, robust	Less interpretable	Best for this project
Neural Network	Handles complex patterns	Needs large data, complex tuning	Not required here
XGBoost	Very high accuracy	More complex, needs tuning	Suitable for large-scale systems

```
PS C:\Users\Praneeth\Desktop\L&T\week1> py predict.py
Test Case 1: [25, 50, 80, 1012] → Safe
Test Case 2: [30, 65, 180, 1010] → Safe
Test Case 3: [38, 85, 400, 1005] → Moderate
Test Case 4: [22, 40, 60, 1020] → Safe
Test Case 5: [35, 75, 250, 1008] → Moderate
PS C:\Users\Praneeth\Desktop\L&T\week1>
```

7. Model Evaluation

The model performance was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

Results:

- Accuracy: **98.7%**
- High precision and recall across all classes
- Minimal misclassification

```
PS C:\Users\Praneeth\Desktop\L&T\week1> py advanced_model.py
Fitting 3 folds for each of 4 candidates, totalling 12 fits

Best Parameters: {'learning_rate': 0.1, 'max_depth': 6, 'n_estimators': 100}

Accuracy: 0.9963

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     17614
           1       1.00      1.00      1.00     17804
           2       0.99      0.99      0.99      4582

    accuracy                           1.00     40000
   macro avg       1.00      1.00      1.00     40000
weighted avg       1.00      1.00      1.00     40000


Confusion Matrix:
[[17574    40     0]
 [   42 17732    30]
 [    0    36  4546]]

Model & scaler saved ✅

Generating SHAP summary plot...

Generating SHAP force plot for one sample...
```

8. Prediction and Model Saving

The trained model was tested using new input data and successfully predicted the corresponding risk level. The model was saved as a .pkl file using joblib, enabling reuse for real-time prediction and system deployment.

9. Outcome

At the end of this milestone, a complete pipeline was successfully developed, including:

- Synthetic dataset generation
- Data preprocessing
- Risk classification
- Machine learning model training and evaluation

This milestone provides a strong foundation for integrating machine learning predictions with real-time IoT data and visualization dashboards in future stages.